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3.2.5 In practice, MGS is more accurate

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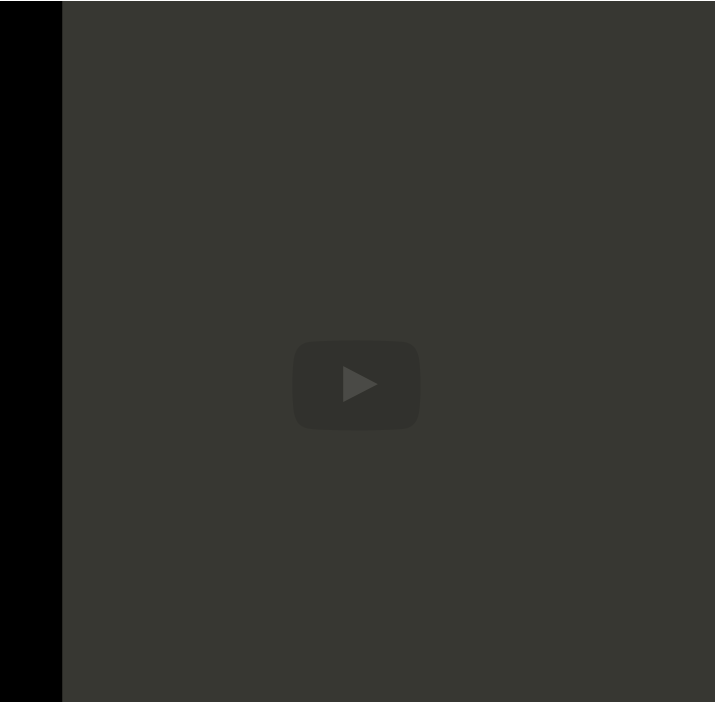
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method. But we know that computers don't compute exactly. We store real numbers as floating-point numbers and in doing so we incur some round-off error. Also when we compute with these floating-point numbers we may incur some more round-off error. So a question is, "Do these methods actually compute mutually orthonormal columns from a given matrix A?" In the next homework, you're going to

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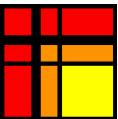
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Reading assignment

1/1 point (graded)



Read Unit 3.2.5 of the notes. [\[LINK\]](#)

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Homework 3.2.5.1

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Let $\epsilon = \sqrt{\epsilon_{\text{mach}}}$ and consider the matrix

$$A = \left(\begin{array}{c|c|c} 1 & 1 & 1 \\ \epsilon & 0 & 0 \\ 0 & \epsilon & 0 \\ 0 & 0 & \epsilon \end{array} \right) = \left(\begin{array}{c|c|c} a_0 & a_1 & a_2 \end{array} \right).$$

By hand, apply both the CGS and the MGS algorithms with this matrix, rounding $1 + \epsilon_{\text{mach}}$ to 1 whenever encountered in the calculation.

Upon completion, check whether the columns of Q that are computed are (approximately) orthonormal.

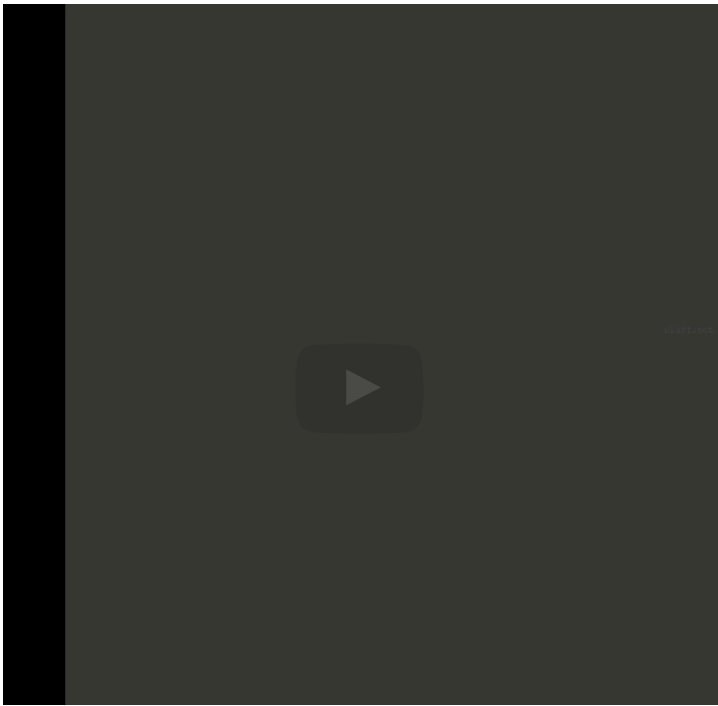
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If you then subtract out the component [that actually] in the direction of q_1 that actually is in the original a_2 , then you're left with an error which may have the relatively large component in the direction of q_1 . And therefore the q_2 that you compute is actually not particularly orthogonal to q_1 . Now if you contrast this with what happens in Modified Gram Schmidt method, ..

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